

José **Jácome**

Mechatronics M.Sc. · Computer Vision Engineer

Santiago de Píllaro, Ecuador

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"Mechatronics Engineer with experience in Mechatronics, Embedded Systems and Computer Vision. I worked as a Computer Vision Engineer in ARS Traffic & Transportation Technology. Linux user and competent Python/C++ programmer with previous academic research experience. Interested in work in R&D in Mechatronics, Computer Vision or IoT with embedded systems. Available for relocation."

Education

University of Debrecen

Debrecen, Hungary

Mechatronical Engineering M.Sc. | GPA: 3.5 | EQF Level 7 — *Sept. 2019 - June 2021*

- Stipendium Hungaricum Scholarship Holder
- TalentUD University of Debrecen Honours College Participant – 2021(Hungary) *[Link]*
- **Thesis:** Multi-Object Tracking for Traffic Monitoring System *[Code] [Video]*

University of the Armed Forces

Latacunga, Ecuador

Mechatronics Engineering B.Sc. | GPA: 3.7 | EQF Level 6 — *Sept. 2011 - Jan. 2018*

- **Thesis:** Development of a Weed Removal Robot *[Code] [Video]*

Skills

C++ Programming Language	STL, OpenCV, QT, Boostlib, GStreamer, Nvidia DeepStream SDK, CMake, VCPKG, Conan
Python Programming Language	Tensorflow, Matplotlib, MQTT, Triton Client, OpenVINO, Numpy, Jupyter Notebooks, Pandas, FastAPI
DevOps	GIT, Docker, Jira, Open API, Triton Server, Bash scripting, valgrind, pytest, GitHub Actions
Computer Vision	Geometrical and Filtering Operations, Image Classification, Object Detection, Instance Segmentation, Dataset Creation/Validation/Wrangling
Mechatronics Engineering	Mechatronics Design, Instrumentation and Signal conditioning, PCB Design, Electrical and Pneumatic Automation, Robotics
Embedded Systems	MPLab X, Arduino IDE, PICC, ESP32, Microchip, Nvidia Jetson, ONNX Inference, I2C, UART, SPI, Ethernet
Computer-aided design	Solidworks, Autodesk Inventor, FreeCad, KiCAD
Languages	Spanish, English TOEFL iBT *[89/120]*, Hungarian Level *[KER A1.1.]*

Experience

ARS Traffic and Transportation Technologies B.V.

The Hague, Netherlands (Online)

Computer Vision Engineer — *Apr. 2022 - Present*

- **Project's Participation:** 🇳🇱 Groningen Parking Enforcement (2026), 🇳🇱 NDW Highway Monitoring (2024-2025), 🇱🇻 Vidzeme Region Traffic Monitoring (2023), 🇳🇱 Railroad Safety Enforcement (2022).
- Developed custom end-to-end computer vision applications for vehicle segmentation, detection, classification, and ANPR across 200+ critical locations.
- Architected high-performance C++20 and Python GStreamer pipelines to process real-time, multi-stream video feeds.
- Trained and fine-tuned deep learning models (PyTorch & Keras) for instance segmentation, object detection, vehicle classification, and character recognition.
- Quantized and optimized models for low-latency roadside edge computing on Nvidia Jetson, Intel, and Google Coral TPUs.
- Scalable Computer Vision APIs with FastAPI, OpenAPI, OpenVINO, Docker, and Valkey.
- Designed Docker-based microservices and automation shell scripts for cross-platform deployment on Linux and Windows machines.

IDX Sistemas

Ambato, Ecuador

Mechatronics & Artificial Intelligence Engineer — *Jun. 2018 - Jan. 2021*

- Development of tailor-made mechatronics, artificial intelligence, and software solutions.
- Mechanical and electronic design with FreeCAD and KiCad.
- Build physical prototypes and software demos.
- Assistance in realization of Artificial Intelligence Software projects.
- Documentation and project's presentation.

Research Activities

2019	**Cyber-physical system for quality control of spur gears through artificial vision techniques.**, JÁCOME RAMÍREZ, Fernando Israel; JÁCOME BARRIONUEVO, José Misael. In 2019 IEEE Fourth Ecuador Technical Chapters Meeting (ETCM). IEEE, 2019. p. 1-6.	*Ecuador*
2017	**Precise weed and maize classification through convolutional neural networks.**, CÓRDOVA-CRUZATTY, Andrea; BARRENO BARRENO, Mauricio Daniel; JÁCOME BARRIONUEVO, José Misael. In 2017 IEEE Second Ecuador Technical Chapters Meeting (ETCM). IEEE, 2017. p. 1-6	*Ecuador*

Portafolio

2026	**Happy Paws Píllaro Django Web Application**, Available on *[GitHub]*
2025	**ESP32 based GPS Cycling Tracker**, Available on *[GitHub]*
2025	**Web Python for Management and Statistics of Low-Cost Neutering Campaigns**, Available on *[GitHub]*
2022	**DIY C Language Seven Segment based Electronic Marker for Volleyball**, Available on *[GitLab]*
2021	**Lightweight People Counter with Computer Vision**, Available on *[YouTube]*
2021	**Open-Source Traffic Monitoring and Situation Recognition System**, Available on *[GitLab]*
2021	**Project DIALUNG – Software for detection of pulmonary diseases - ASEIB**, Available on *[GitHub]*
2021	**Workshop AI on the Edge and Performance Optimization - ESPE Sede Latacunga**, Available on *[GitHub]*
2019	**Tello Drone following system via Computer Vision**, Available on *[GitLab]*

Honors & Awards

International Awards

2021	**Gold Award**, Outstanding Performance in Scientific Student's Association *[Link]*	*Debrecen, Hungary*
2020	**1st place**, Conference of Scientific Students' Association Faculty of Engineering *[Link]*	*Debrecen, Hungary*

Domestic Awards

2022	**1st Place**, 1st Prize, Datajam: Data analysis for water conservation – Datalat Foundation	*Quito, Ecuador*
2017	**3rd place of Electric Control**, Festival of Knowledge and Skills - WordSkill Ecuador UDLA	*Quito, Ecuador*
2017	**3rd place**, 11th Robotic Contest ESPE Latacunga - Innovation Project	*Latacunga, Ecuador*
2016	**1st place**, 10th Robotic Contest ESPE Latacunga- Minisumo Battlebots	*Latacunga, Ecuador*

Community

2025	**Fourth Youth's Award**, For Outstanding Social Activies, Foundation Red Juvenil Píllaro	*Santiago de Píllaro, Ecuador*
2024 - Present	**Member of Dogs and Cats' Welfare Foundation**, *Happy Paws Píllaro*	*Santiago de Píllaro, Ecuador*

Certificates

2021	**Workshop AI on the Edge and Performance Optimization**, ESPE Sede Latacunga. Available on *[Link]*
2021	**AI on the Edge and Performance Optimization**, ESPE Sede Latacunga. Available on *[Link]*
2021	**35th National Scientic Student Conference Participation**, OTDK. Available on *[Link]*
2020	**Intel® Edge AI for IoT Developers Nanodegree**, Udacity. Available on *[Link]*
2018	**ROS1x: Hello (Real) World with ROS - Robot Operating System**, DelftX|edX. Available on *[Link]*
2018	**IEEE Ecuador Technical Chapters Meeting (ETCM 2017),**, ETCM IEEE. Available on *[Link]*

Presentation

2nd Congress of Science and Technology ESPE 2021 with focus on Mechatronics and Automotive Engineering — *Latacunga, Ecuador*

Presenter for <Artificial Intelligence for Edge Devices> — *Sept. 2021*

- Brief introduction of Artificial Intelligence for Mechatronics Applications.
- Techniques to run AI models on edge devices for remote operation.